A Quick Look at Image Processing with Deep Learning

Social media networks like Facebook have a large user base and an even larger accumulation of data, both visual and otherwise. Face recognition is an important feature of such sites, and has been made possible by deep learning. Let's find out how.

Deep learning is a type of machine learning in which a model learns to perform classification tasks directly from images, text or sound. Deep learning is usually implemented using neural network architecture. The term *deep* refers to the number of layers in the network—the more the layers, the deeper the network. Traditional neural networks contain only two or three layers, while deep networks can have hundreds.

In Figure 1, you can see that we are training the system to classify three animals – a dog, a cat and a honey badger.

A few applications of deep learning

Here are just a few examples of deep learning at work:

- A self-driving vehicle slows down as it approaches a pedestrian crossing.
- An ATM rejects a counterfeit bank note.
- A smartphone app gives an instant translation of a street sign in a foreign language.

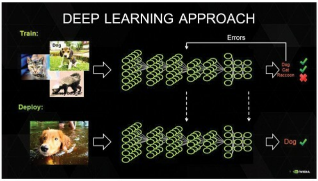

Figure 1: The deep learning approach

Deep learning is especially well-suited to identification applications such as face recognition, text translation, voice recognition, and advanced driver assistance systems, including lane classification and traffic sign recognition.

The learning process of deep neural networks

A deep neural network combines multiple non-linear processing layers, using simple elements operating in parallel. It is inspired by the biological nervous system, and consists of an input layer, several hidden layers, and an output layer. The layers are interconnected via nodes, or neurons, with each hidden layer using the output of the previous layer as its input.

How a deep neural network learns

Let's say we have a set of images, where each image contains one of four different categories of objects, and we want the deep learning network to automatically recognise which

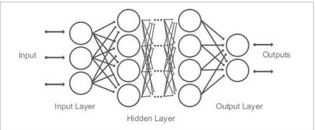

Figure 2: Deep learning neural networks

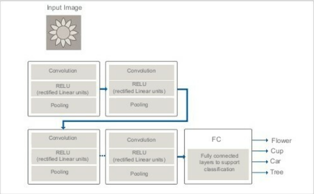

Figure 3: Neural network data training approach